

VINAYAK KOLI

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EDUCATION

Indraprastha Institute of Information Technology Delhi (IIIT-D)

2023 – 2027

Bachelor of Technology in Computer Science and Social Sciences

Delhi, India

Coursework: Database Management Systems, Object-Oriented Programming, Advanced Programming, Operating Systems, Data Structures and Algorithms, Algorithm Design, Collaborative Filtering

SKILLS

Languages: Python, Java, C++, JavaScript, TypeScript, SQL

Frameworks and Libraries: React Native, FastAPI, PyTorch, LightGBM, LangChain, HuggingFace Transformers, Gradio

Tools and Platforms: Git, Appwrite, Expo, Optuna, Chrome Manifest V3, Vite, Zustand, Postman, VS Code, Android Studio

EXPERIENCE

AI Model Evaluation Research

Jan 2026 – Present

Undergraduate Researcher under Prof. Suman Roy / IIIT Delhi

- Designed and implemented a custom evaluation system for the DeepMind Code Contests dataset, enabling automated execution, correctness checking, failure detection, and structured result logging at scale.
- Benchmarked Small Language Models (SLMs) on DeepMind Code Contests and HumanEval datasets across 165+ problems, analysing pass@k, execution accuracy, and multi-step reasoning failure modes.
- Contributing to ongoing research on AI automation using SLMs for multi-step code generation and agentic reasoning workflows.

Multi-Agent SOAP Note Generation (Ongoing)

Apr 2026 – Present

Undergraduate Intern under Prof. Suman Roy / IIIT Delhi

- Designing a manager-specialist multi-agent LLM architecture to automate clinical reasoning and SOAP note generation from doctor-patient conversations; implementing iterative agent debate for consensus-based diagnosis.
- Exploring multimodal inputs (text and clinical images) with Retrieval-Augmented Generation (RAG) to reduce hallucinations; applying weakly supervised learning for structured medical note generation with limited annotated data.
- Evaluating outputs using clinical metrics (precision, recall, F1, semantic alignment); investigating failure modes such as groupthink and bias amplification in multi-agent collaborative reasoning.

PROJECTS

Hybrid Movie Recommendation System | Python, PyTorch, LightGBM, LoRA, Optuna, Gradio | [GitHub](#)

- Designed an end-to-end multi-stage recommendation pipeline on MovieLens-1M (1M+ ratings, 6,035 users) with a 7-model parallel ensemble (BPR, LightGCN, NeuMF, SASRec, EASE, DCN, Popularity) and rank-normalised feature fusion.
- Fine-tuned Gemma-2 (9B) with Low-Rank Adaptation (LoRA) on a structured yes/no preference dataset; extracted Yes-token logit probabilities as orthogonal semantic features; automated hyperparameter optimization via Optuna TPE over 8 parameters with SQLite-persisted studies.
- Achieved Hit Rate at 10 (HR@10) = 0.797, NDCG@10 = 0.636, 78% catalog coverage, and personalisation score = 0.99 on a frozen leave-one-out evaluation protocol with 99 negatives per user across 6,035 users.

Vocal for Local (VoLo) | React Native (Expo), FastAPI, Appwrite, Python | [GitHub](#) | [LinkedIn](#)

- Built a cross-platform mobile app (14 screens, Expo Router) with a FastAPI and Pydantic v2 backend; implemented 11 REST endpoints across 3 resource domains with ownership-based authorization (HTTP 403 guards) and Google OAuth 2.0 via Appwrite.
- Implemented Haversine geodesic distance from scratch, Google Places Autocomplete, and multi-field full-text search with server-side deduplication; executed zero-downtime Appwrite v1.8 TablesDB migration with a field-normalization compatibility layer.
- Improved app performance by approximately 35% via in-memory server-side caching (60-second TTL with prefix invalidation), debounced search, lazy loading, and AbortController timeouts on all fetch calls.

ContextLens | React 18, TypeScript, Chrome MV3, Vite, Zustand, IndexedDB | [GitHub](#)

- Built a Manifest V3 Chrome Extension with an inline floating UI using React and Shadow DOM for complete CSS isolation (zero style leakage) on any webpage; supports contextual AI actions: Summarize, Define, and Ask.
- Engineered a type-safe message-passing system between Content Script, Service Worker, and Side Panel; built a multi-provider AI abstraction layer supporting OpenAI, Anthropic, and local Ollama for zero-cost offline execution.
- Integrated IndexedDB-backed persistent note storage anchored to DOM XPath with XSS-hardened input handling and Zustand global state; achieved zero main-thread blocking by offloading all LLM and database operations to a background Service Worker.

ACHIEVEMENTS AND ACTIVITIES

- 3rd Place** – DevHaven 2.0 Hackathon.
- Captain of Table Tennis Team for Karwaan 2024; Member of Sports Council Operations Team.
- Led a 10+ member team delivering a cancer awareness workshop impacting 100+ individuals at Tihar Jail.